

Python for System Administrators (PYTHOPS)

Course Outline

Duration: 4 Days (32 Hours) | Instructor-led with Hands-on Labs

Module 1: Python Fundamentals for Administrators

- Python basics, syntax, variables, data types
- Control statements and loops
- Functions and modules

Lab: Basic administration scripts

Module 2: Working with Files and Directories

- File read/write operations
- CSV and text processing
- Directory management and permissions Lab:

Log analysis and backup automation

Module 3: Operating System Automation

- OS, Pathlib and Shutil modules
- Process management
- Environment variables and shell commands

Lab: Process monitoring and service status checker

Module 4: System Administration Using Python

- System information gathering
- CPU, memory and disk monitoring
- User and group administration concepts Lab:

Server health check utility

Module 5: Log Management and Monitoring

- Log parsing
- Regular expressions
- Alert generation

Lab: Error log parser

Module 6: Network Automation

- DNS and IP utilities
- Socket programming basics
- Connectivity testing

Lab: Ping automation and port scanner

Module 7: Process and Service Automation

- Service automation
- Task scheduling
- Exception handling and logging

Lab: Service restart automation

Module 8: Reporting and Capstone Project

- CSV reporting
- Email notifications
- Integrated automation solution

Project: Server Health Monitoring System

Lab Environment Requirements

- Windows 11
- 8/16 GB RAM, 2 CPU Cores, 30 GB disk
- Python 3.11+, VS Code, Git
- Packages: psutil, pandas, requests, schedule
 - Local admin/sudo access

General - NIIT Data